

Pierre Moreau

PhD in Computer Science · Data Scientist / ML Engineer — AI & Healthcare

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Summary

PhD in Computer Science (AI / Machine Learning) specialised in healthcare, combining applied research and engineering. Skilled in multimodal learning, generative AI (LLMs, RAG) and clinical data processing. Committed to building interpretable, reliable AI for real-world clinical impact.

Skills

Languages: Python, SQL, JavaScript, Bash, PHP, HTML/CSS

ML & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, XGBoost

LLMs & Generative AI: Fine-tuning (PEFT, LoRA), RAG (LangChain), prompting, Whisper (speech-to-text), OCR

Vision & NLP: Segmentation (UNetR, SwinUNet), OpenCV, Pillow, NLTK

Data Science: NumPy, Pandas, SciPy, Matplotlib, Seaborn ; PCA, DTW, ANOVA, Chi²

MLOps & Deployment: FastAPI, Django, Flask, Docker, Git

Responsible AI: Explainability (SHAP, LIME), bias/fairness analysis, GDPR & AI Act

Tools: Git, Docker, VS Code / Cursor, Jupyter, Claude Code / Codex

Professional Experience

Post-doctoral Researcher — AI & Healthcare, University of Picardie Jules Verne — Amiens Sept 2025 – present

- RAUC project: designing an optimised care pathway for digestive surgical emergencies, integrating AI and e-health, aimed at predicting post-operative complications.

Data & AI Engineer — AI & Data Science Lead (Healthcare), Capgemini Engineering – Paris Mar 2024 – May 2025

- Designed an AI-based medical assistant demonstrator with multimodal workflows.
- Built NLP pipelines: OCR of blood test results, speech-to-text transcription (Whisper), automatic keyword extraction to enrich metadata.
- Fine-tuned LLMs (OpenAI, Mistral) with Hugging Face Transformers and PEFT for clinical summaries and audio transcription.
- Developed a multimodal RAG pipeline (LangChain) synthesising structured and unstructured medical data.
- Applied research on placenta segmentation (UNetR, SwinUNet) to improve accuracy in prenatal imaging.
- Coordinated AI researchers, data scientists and clinicians; project and team management (15-20 people).

Teaching & Research Assistant (ATER), University of Picardie Jules Verne – Amiens Sept 2022 – Aug 2023

- Tutorials and lab sessions for undergraduates (years 1-3) — PHP, Bash, C, advanced programming.

PhD Candidate — Doctoral Research, University of Picardie Jules Verne – Amiens Sept 2019 – June 2023

- Research on human motion modelling and analysis (in collaboration with Amiens University Hospital); built a comprehensive motion database.
- Analysis (PCA, Dynamic Time Warping, correlation) and optimisation via genetic algorithms for feature selection.
- Deep learning modelling (RNN, LSTM, CNN, residual networks, FcNN) and SVM; interpretable biomechanical representation using matrix formulations.

IT Engineer / Developer (apprenticeship then permanent), LV Informatique – Amiens Sept 2017 – Sept 2019

- Developed custom ERP software (Lynéade then Lynéa2, WinDev): business processes (accounting, purchasing, sales, inventory, after-sales), SQL database administration, client support and customisation.

Education

University of Picardie Jules Verne, PhD in Computer Science — Artificial Intelligence – Amiens Sept 2019 – June 2023

- Thesis: analysis of human motion using artificial intelligence tools, in collaboration with Amiens University Hospital.

University of Picardie Jules Verne, MSc in Computer Methods Applied to Business Management (MIAGE) – Amiens Sept 2017 – June 2019

University of Picardie Jules Verne, BSc in Computer Methods Applied to Business Management (MIAGE) – Amiens Sept 2014 – June 2017

Publications

A Genetic Algorithm-Optimized PID Controller for Robust Tumor Treatment via Combined Immuno-Chemotherapy 2026
Jérôme Bosche, Consibelle Djiffouo Dzoyim, *Pierre Moreau*
(International Conference on Control, Decision and Information Technologies (CoDIT) — accepted)

Towards Better Placenta Segmentation for Earlier Preeclampsia Detection on 3D Doppler Images 2025
Charlotte Alliod, Emilien Micard, *Pierre Moreau*, Léo Challier, et al.
(Computerized Medical Imaging and Graphics (CMIG) — submitted)

Two-branch neural network using two data types for human activity recognition 2023
Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(IEEE Sensors Journal)

A motion recognition technique based on linear matrix representation to improve Parkinson's disease treatments 2021
Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(9th International Conference on Systems and Control (ICSC))

A motion recognition algorithm using polytopic modeling 2020
Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(7th International Conference on Control, Decision and Information Technologies (CoDIT))

Certifications

Generative AI with Large Language Models — DeepLearning.AI & AWS (2024)
Building Multimodal Search and RAG — DeepLearning.AI (2025)
Build Basic Generative Adversarial Network (GAN) — Coursera (2025)
Python NLTK for Beginners — Coursera (2024)